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NARULA INSTITUTE OF TECHNOLOGY
An Autonomous Institute under MAKAUT
2025
Odd Semester - 2025-26
M(IT)101 - ENGINEERING MATHEMATICS - I

TIME ALLOTTED: 3Hours

Maximum Marks : 70

Instructions to the candidate:*Figures to the right indicate full marks.**Draw neat sketches and diagram wherever is necessary.**Candidates are required to give their answers in their own words as far as practicable***Part A****Answer any ten from the following, choosing the correct alternative of each question: 10×1=10**

1. The Product of any matrix with its transpose is - (1) C01 BL1 PO1
(a) symmetric (b) skew-symmetric
(c) orthogonal (d) none.
2. Lagrange's MVT is obtained from Cauchy's MVT for the function (1) C03 BL4 PO2
 $\phi(x)$ and $\psi(x)$ by putting $\psi(x) = ?$.
(a) 0 (b) 1
(c) x (d) none
3. If $x=r\cos\theta$, $y=r\sin\theta$, then $\frac{\partial(x,y)}{\partial(r,\theta)}$ is - (1) C04 BL3 PO2
(a) $-\frac{1}{r}$. (b) $-r$.
(c) r . (d) 0
4. The value of $\int_1^0 \int_0^1 (x+y) dx dy$ is - (1) C02 BL3 PO1
(a) 2 (b) 3
(c) -1 (d) 0
5. The critical point of the function $f(x,y)=xy$ is (1) C01 BL1 PO1
(a) (1,1) (b) (1,-1)
(c) (-1,1) (d) (0,0)

6. The system of linear equations $AX = B$, where A is an $n \times n$ matrix, has (1) C01 BL1 PO1
no solution or infinitely many solutions if and only if
- (a) $\det A \neq 0$. (b) $\det A = 1$.
(c) $\det A = 0$. (d) $\det A < 0$.
7. The eigenvalues of the matrix $\begin{pmatrix} 5 & 2 \\ 0 & 3 \end{pmatrix}$ are (1) C02 BL2 PO1
- (a) 5, 2. (b) 5, 0.
(c) 3, 0. (d) 3, 5.
8. Which of the following functions does not satisfy the conditions of (1) C02 BL1 PO1
Rolle's Theorem on $[-1, 1]$?
- (a) $f(x) = x^2$ (b) $f(x) = \sin x$
(c) $f(x) = \tan x$ (d) $f(x) = x^4 + 1$
9. The series $1 + r + r^2 + r^3 + \dots$, for $|r| < 1$ converges to (1) C01 BL1 PO1
- (a) r^2 (b) r^3
(c) $1 / (1 - r)$ (d) 0
10. The volume of the tetrahedron bounded by the plane $x+y+z=1$ and the (1) C02 BL2 PO1
co-ordinate planes is
- (a) $1/2$ (b) $1/3$
(c) 1 (d) $1/6$
11. The Jacobian of the two functions $f(x, y) = x$ and $g(x, y) = xy$ is (1) C02 BL2 PO1
- (a) 0 (b) y
(c) x (d) 1
12. If C is circle $x^2 + y^2 = 4$ then $\int_C x^2 dx =$. (1) C02 BL2 PO1
- (a) 0 (b) 3
(c) 1 (d) $1/3$

Part B**(Answer any three of the following) 3x5=15**

13. Find the rank of the matrix $A = \begin{pmatrix} 1 & 2 & 1 \\ -1 & 0 & 2 \\ 2 & 1 & -3 \end{pmatrix}$. (5) C03 BL2 PO2
14. Is the Rolle's theorem applicable to the following function (5) C02 BL4 PO2
 $f(x) = 1 - x^{\frac{2}{3}}$ in $[-1, 1]$?

15. Check wheather the funtion

(5) CO2 BL3 PO1

$$f(x, y) = \begin{cases} \frac{2xy}{x^2+y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$$

is continuos or not at the origin.

16. Evaluate
- $\int \int_R y \, dx \, dy$
- where R being the region bounded by the parabola
- $y^2 = 4x$
- and
- $x^2 = 4y$
- .

(5) CO4 BL6 PO1

17. Determine the conditions for which the systems

(5) CO3 BL3 PO1

$$x + 4y + 2z = 1$$

$$2x + 7y + 5z = 2a$$

$$4x + by + 10z = 2a + 1$$

has (i)no solution, (ii)unique solution.

Part C

(Answer any three of the following) 3x15=45

18. (a) Verify Rolle's theorem for the function in the specified interval:

(5) CO1 BL1 PO1

$$f(x) = e^x \cos(x) \text{ in } \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

- (b) Using Mean Value Theorem to prove that

(5) CO3 BL4 PO1

$$\frac{2x}{1-x^2} > \log\left(\frac{1+x}{1-x}\right) > 2x, 0 < x < 1$$

- (c) State Taylor's Series expansion (Infinite from).

(5) CO3 BL3 PO1

Using it expand $f(x) = \sin x$ in powers of x on an infinite series stating the conditions under which the expansion is valid.

19. (a) If
- $u = \sin^{-1} \frac{x+2y+3z}{\sqrt{x^8+y^8+z^8}}$
- , show that

(7) CO4 BL6 PO2

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} + 3 \tan u = 0$$

- (b) Examine the function
- $f(x, y) = x^3 + y^3 - 3axy$
- for maxima and minima.

(5) CO3 BL3 PO1

- (c) The value of $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2}$ if its exists (3) C03 BL3 PO3
20. (a) Find the extrema and the saddle points of $f(x,y) = 4x^2 + 4y^2 + x^3y + xy^3 - xy - 4$. (8) C03 BL3 PO1
- (b) Expand $\cos x$ in power of x , in infinite series. (7) C03 BL3 PO1
21. a) Evaluate $\int_C ydx + xdy$ where C is the curve $y = x^2$ from (0,0) to (1,1). (7) (15) C03 BL3 PO1
- b) Evaluate the volume integral $\int \int \int_V (x + 2y + 3z) dV$ where V is the cube bounded by $0 \leq x \leq 1, 0 \leq y \leq 1, 0 \leq z \leq 1$. (8)
22. a) Use Cayley-Hamilton Theorem to find A^{100} , where (15) C04 BL5 PO1
- $$A = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \quad (6)$$
- b) Check whether the matrix is Diagonalisable and calculate the Diagonal matrix
- $$A = \begin{pmatrix} 3 & 2 & 1 \\ 2 & 3 & 1 \\ 0 & 0 & 1 \end{pmatrix} \quad (9)$$